Misr University for Science & Technology
Faculty of Engineering

CSE491

Computer Interface

Ultrasonic Distance Sensor

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1. Project Objective

The objective of this project is to design and implement an ultrasonic distance measurement system using a PIC microcontroller and the HC-SR04 ultrasonic sensor. The measured distance is displayed in real time on a 16×2 LCD.

2. Components Used

- PIC Microcontroller (PIC16F887 / PIC16F877A)

- HC-SR04 Ultrasonic Sensor

- 16×2 LCD Display (4-bit mode)

- 8 MHz Crystal Oscillator

- Potentiometer for LCD contrast

- Resistors and Capacitors

- 5V Regulated Power Supply

3. Working Principle

The HC-SR04 ultrasonic sensor operates at a frequency of 40 kHz. A 10 µs trigger pulse is sent from the microcontroller to the sensor. The sensor emits an ultrasonic burst which reflects from an object and returns back to the sensor. The Echo pin outputs a pulse whose width represents the round-trip travel time of the ultrasonic wave.

4. Speed of Sound

The speed of sound in air at 25°C is approximately 343 m/s, which is equivalent to 0.0343 cm/µs.

5. Distance Calculation Formula

Distance (cm) = Time (µs) / 58
The division by 2 is required because the measured time represents the forward and return journey.

6. Timing Characteristics

- Trigger pulse width: 10 µs
- Minimum measurable distance: ~2 cm
- Maximum measurable distance: ~4 meters

7. Temperature Effect

The speed of sound changes with temperature according to the formula:
$c = 331 + 0.6 \times T\ (°C)$
An increase in temperature slightly increases the measured distance.

8. Sources of Error

- Temperature variations
- Angled or soft surfaces
- Electrical noise
- Long connecting wires
- Cross-talk from nearby ultrasonic sensors

9. Methods to Increase Accuracy

- Using timer capture modules
- Averaging multiple readings
- Calibration using offset values
- Using short and well-shielded wires
- Adding bypass capacitors to power supply

10. Benefits of the Project

- Non-contact distance measurement
- High accuracy and fast response
- Low cost and easy implementation
- Real-time measurement capability
- Suitable for educational and industrial use

11. Applications

- Obstacle detection
- Robotics and automation
- Parking assistance systems
- Industrial distance monitoring
- Smart sensing systems

12. Code

```
/ ======================================
// Ultrasonic Distance Meter
// PIC16F877A + HC-SR04 + LCD 16x2
// Crystal: 8 MHz
// Compiler: MikroC PRO for PIC
// ======================================

// -------- LCD pin definitions ----------
sbit LCD_RS at RD2_bit;
sbit LCD_EN at RD3_bit;
sbit LCD_D4 at RD4_bit;
sbit LCD_D5 at RD5_bit;
sbit LCD_D6 at RD6_bit;
sbit LCD_D7 at RD7_bit;

sbit LCD_RS_Direction at TRISD2_bit;
sbit LCD_EN_Direction at TRISD3_bit;
sbit LCD_D4_Direction at TRISD4_bit;
sbit LCD_D5_Direction at TRISD5_bit;
sbit LCD_D6_Direction at TRISD6_bit;
sbit LCD_D7_Direction at TRISD7_bit;
// ---------------------------------------

unsigned int distance;
char txt[7];

void main() {
```

```
// -------- I/O Configuration ----------
TRISB = 0b00010000;   // RB4 = Echo (input), RB0 = Trig (output)
PORTB = 0x00;

TRISD = 0x00;         // LCD on PORTD
PORTD = 0x00;
// -------------------------------------

// -------- LCD Initialization --------
Lcd_Init();
Lcd_Cmd(_LCD_CLEAR);
Lcd_Cmd(_LCD_CURSOR_OFF);

Lcd_Out(1,1,"Ultrasonic");
Lcd_Out(2,1,"Distance Meter");
Delay_ms(2000);
Lcd_Cmd(_LCD_CLEAR);
// -------------------------------------

// -------- Timer1 Configuration -------
// Fosc = 8 MHz
// Fosc/4 = 2 MHz
// Prescaler = 1:2  → Timer tick = 1 µs
T1CON = 0x10;        // T1 prescaler 1:2, Timer1 OFF initially
// -------------------------------------

while(1) {

   // Clear Timer1
```

```
TMR1H = 0;
TMR1L = 0;

// Send 10 us trigger pulse
PORTB.F0 = 1;
Delay_us(10);
PORTB.F0 = 0;

// Wait for echo to go high
while(!PORTB.F4);

// Start Timer1
T1CON.F0 = 1;

// Wait for echo to go low
while(PORTB.F4);

// Stop Timer1
T1CON.F0 = 0;

// Read timer value
distance = (TMR1H << 8) | TMR1L;

// Convert time (µs) to distance (cm)
// distance(cm) ≈ time / 58
distance = distance / 58;

Lcd_Cmd(_LCD_CLEAR);
```

```
      if(distance >= 2 && distance <= 400) {

         IntToStr(distance, txt);
         Ltrim(txt);

         Lcd_Out(1,1,"Distance:");
         Lcd_Out(2,1,txt);
         Lcd_Out(2,5,"cm");
      }
      else {
         Lcd_Out(1,1,"Out of range");
      }

      Delay_ms(400);
   }
}
```

13. Code explanation

System Overview and Hardware Configuration

1. Microcontroller: PIC16F877A.
2. Crystal Frequency: 8 MHz.
3. Compiler: MikroC PRO for PIC.
4. Ultrasonic Sensor (HC-SR04) Pins:
    - Trigger (Trig): Connected to RB0 (Output). This pin sends the initial $10\ \mu s$ pulse to start the measurement.
    - Echo (Echo): Connected to RB4 (Input). This pin receives the returning pulse, and its duration is measured.
5. LCD 16x2 Pins:
    - The control pins (RS, EN) and data pins (D4-D7) are connected to PORTD (RD2 through RD7).

Timer Configuration for Precision Timing

The program uses Timer1 to measure the pulse width of the Echo signal.

- The instruction cycle frequency is $F_{inst} = F_{osc}/4 = 8\text{ MHz}/4 = 2\text{ MHz}$.
- Timer1 is configured with a 1:2 Prescaler (T1CON = 0x10).
- Timer Tick Period: $\text{Time} = \text{Prescaler} \times (1 / F_{inst}) = 2 \times (1 / 2\text{ MHz}) = \mathbf{1\ \mu s}$.
- This $1\ \mu s$ tick time means the 16-bit value read from Timer1 ($\text{TMR1H} | \text{TMR1L}$) directly represents the total travel time in microseconds (μs).

Distance Measurement Logic (Time-of-Flight)

The core distance measurement process takes place inside the infinite while(1) loop:

1. Reset Timer: Timer1 (TMR1H and TMR1L) is cleared to zero to start a new measurement.
2. Trigger Pulse: A $10\ \mu s$ HIGH pulse is generated on the Trig pin (RB0) to initiate the ultrasonic sensor.
3. Wait for Echo HIGH: The code blocks until the Echo pin (RB4) goes HIGH (while(!PORTB.F4);), signaling the sound pulse has been emitted.
4. Start Timing: Timer1 is started (T1CON.F0 = 1;).
5. Wait for Echo LOW: The code blocks until the Echo pin goes LOW (while(PORTB.F4);), marking the moment the sound has traveled to the object and returned.
6. Stop Timing: Timer1 is stopped (T1CON.F0 = 0;).
7. Read Time: The total time in μs is read from the 16-bit Timer1 registers: distance = (TMR1H << 8) | TMR1L;.

Calculation and LCD Display

The raw time value is converted into distance and displayed on the LCD:

1. Distance Conversion: The measured time T (in μs) is converted to distance D (in cm) using the standard formula for the HC-SR04:

$$D (\text{cm}) \approx \frac{T (\mu s)}{58}$$

This is implemented as: distance = distance / 58;.

2. Range Validation: The calculated distance is checked to be within the HC-SR04's effective range of 2 cm to 400 cm.
3. Display Output:
    - If the distance is in range, the value is converted to a string (IntToStr) and displayed on the LCD as "Distance:" on line 1 and the value followed by "cm" on line 2.
    - If the distance is out of range, the LCD displays "Out of range".
4. Measurement Delay: A 400 ms delay (Delay_ms(400);) is introduced to space out the measurement cycles.

14. Conclusion

This project demonstrates an effective method for distance measurement using ultrasonic sensing. It combines simplicity, accuracy, and low cost, making it suitable for many real-world applications.

Ultrasonic Distance Measurement Using PIC Microcontroller

1. Project Title

Ultrasonic Distance Meter Using PIC16F887 and HC-SR04

2. Objective

The aim of this project is to measure the distance between an object and an ultrasonic sensor and display the measured distance on a 16×2 LCD.

3. Components Used

- PIC16F887 Microcontroller

- HC-SR04 Ultrasonic Sensor

- 16×2 LCD (4-bit mode)

- 8 MHz Crystal Oscillator

- Potentiometer

- 5V Power Supply

4. Working Principle

The ultrasonic sensor sends a pulse and receives the reflected signal. The PIC uses Timer1 to measure the echo duration and calculate distance.

5. Formula Used

Distance (cm) = Time (µs) / 58

6. Applications

- Obstacle detection

- Robotics

- Parking systems

7. Conclusion

This project provides an accurate and simple distance measurement system using an ultrasonic sensor and a PIC microcontroller.

Ultrasonic Distance Measurement System

Components Used and Their Benefits

• PIC Microcontroller (PIC16F887 / PIC16F877A)

Benefit: Acts as the main control unit. It processes the echo signal, performs timing calculations, controls the LCD, and ensures accurate distance measurement.

• HC-SR04 Ultrasonic Sensor

Benefit: Provides non-contact distance measurement using ultrasonic waves. It is low cost, easy to interface, and offers good accuracy over a wide range.

• 16×2 LCD Display (4-bit mode)

Benefit: Displays the measured distance clearly to the user. 4-bit mode reduces the number of required microcontroller pins, saving hardware resources.

• 8 MHz Crystal Oscillator

Benefit: Provides a stable and accurate clock signal for the microcontroller, ensuring precise timing and reliable delay calculations.

• Potentiometer (LCD Contrast Control)

Benefit: Allows adjustment of LCD contrast for clear and readable display under different lighting conditions.

• Resistors

Benefit: Used for current limiting and signal stabilization to protect components and ensure proper circuit operation.

• Capacitors

Benefit: Help in noise filtering and power supply stabilization, improving overall system reliability.

• 5V Regulated Power Supply

Benefit: Provides a constant and safe voltage required for proper operation of the microcontroller, sensor, and LCD.